Problem 7

In [3]:
```
#p and q are ineger parameters
p, q = var('p q')
```

In [4]:
```
#define the matrix A
A = Matrix(SR, 4, [1, 1, 1, 1, 1, p, p^2, p^3, 1, q, q^2, q^3, 1, p+q,
(p+q)^2, (p+q)^3]); A
```

Out[4]:
```
[        1         1         1         1]
[        1         p       p^2       p^3]
[        1         q       q^2       q^3]
[        1     p + q (p + q)^2 (p + q)^3]
```

In [5]:
```
#factorize the determinant of A
factor(det(A))
```

Out[5]:
```
-(p + q - 1)*(p - q)*(p - 1)*p*(q - 1)*q
```

In [6]:
```
#find the constraints on p and q for wich the matrix A is non-invertible
solve((det(A) == 0),(p,q))
```

Out[6]:
```
[[p == r1, q == 0],
 [p == 0, q == r2],
 [p == 1, q == r3],
 [p == r4, q == -r4 + 1],
 [p == r5, q == r5],
 [p == r6, q == 1]]
```

In [7]:
```
#p and q are the last digits of my ID (std172337), therefore p = 3 and q = 7
p, q = var('p q')
A37 = A.subs({p:3,q:7})
print("If p = 3 and q = 7 then the matrix A is:")
print(A37)
#bring A to its reduced row echelon form
print("\nThe reduced echelon form of this matrix is:")
A37.rref()
```

Out[7]:
```
If p = 3 and q = 7 then the matrix A is:
[   1    1    1    1]
[   1    3    9   27]
[   1    7   49  343]
[   1   10  100 1000]

The reduced echelon form of this matrix is:

[1 0 0 0]
[0 1 0 0]
[0 0 1 0]
[0 0 0 1]
```

In [8]:
```
#compute the rank of A
A37.rank()
```

Out[8]: 4

Problem 8

In [9]:
```
#define the matrices A, B and define vectors x, y
A = matrix([[1,2],[0, 1],[2, 3]])
B = matrix([[1,3,2],[2,5,1],[4,1,0]])
x = vector(SR,var('x1 x2'))
y = vector([25, 15])

print("The matrix A:")
print(A)

print("\nThe matrix B:")
print(B)

print("\nVector x:")
print(x)

print("\nVector y:")
print(y)
```

Out[9]:
```
The matrix A:
[1 2]
[0 1]
[2 3]

The matrix B:
[1 3 2]
[2 5 1]
[4 1 0]

Vector x:
(x1, x2)

Vector y:
(25, 15)
```

In [10]:
```
#compute BB^T and check if it is singular
BBT = B*B.T
print("Matrix BB^T is:")
print(B*B.T)

detBBT = BBT.det
if detBBT == 0:
    print("\nBBT is singular")
else:
    print("\nBBT is not singular")
```

Out[10]:
```
Matrix BB^T is:
[14 19  7]
[19 30 13]
[ 7 13 17]

BBT is not singular
```

In [11]:
```
#compute C=A.T*(BB.T)^(-1)*A
C = A.T*(B*B.T).inverse()*A
```

```
C
```

Out[11]:
```
[   29/25  193/125]
[ 193/125 1306/625]
```

In [12]:
```
#solve the inear system Cx=y in terms of x=(x1, x2)
##using matrix inversion
sol_inv = C.inverse()*y
print("Cx=y with matrix inversion is:")
print(sol_inv)

##using solve function
x1, x2 = var('x x2')
eqs = list(C*vector([x1, x2])-y)
sol_solve = solve(eqs, [x1, x2])
print("\nCx=y with solve function is:")
sol_solve
```

Out[12]:
```
Cx=y with matrix inversion is:
(727, -530)

Cx=y with solve function is:

[[x == 727, x2 == -530]]
```

Problem 9

$$\begin{cases} x+2y-3z=4 \\ 3x-y-5z=2 \\ 4x+y+(k^2-9)=k+5 \end{cases} \qquad A=\begin{vmatrix} 1 & 2 & -3 \\ 3 & -1 & -5 \\ 4 & 1 & k^2-9 \end{vmatrix} \qquad B=\begin{vmatrix} 4 \\ 2 \\ k+5 \end{vmatrix}$$

Let's check the consistency of the system using the determinant:

$$\det(A)=1(-1(k^2-9)-1(-5))-2(3(k^2-9)-4(-5))+(-3)(3\cdot 1-4(-1))=$$
$$=-k^2+9+5-2(3k^2-27+20-3(3+4)=-k^2+14-6k^2+14-21=$$
$$=7-7k^2=7(1-k^2)=\boxed{7(1-k)(1+k)=\det(A)}$$

The system has one solution when $\det(A)\neq 0$

$7(1-k)(1+k)\neq 0 \Rightarrow 1-k\neq 0$ and $1+k\neq 0 \Rightarrow k\neq \pm 1$

Otherwise the system has infinitely many solutions or don't has solutions at all

$7(1-k)(1+k)=0 \Rightarrow 1-k=0$ or $1+k=0 \Rightarrow k=1$ or $k=-1$

Let's use rank method

$k=1$

$4x+y+(k^2-9)=k+5 \Rightarrow 4x+y+(-8)z=1+5 \Rightarrow 4x+y-8z=6$

$$\begin{cases} x+2x-3z=4 \\ 3x-y-5z=2 \\ 4x+y-8z=6 \end{cases} \Rightarrow \begin{pmatrix} 1 & 2 & -3 \\ 3 & -1 & -5 \\ 4 & 1 & -8 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix}=\begin{pmatrix} 4 \\ 2 \\ 6 \end{pmatrix}$$

$$[A_1|B_1]=\left[\begin{array}{ccc|c} 1 & 2 & -3 & 4 \\ 3 & -1 & -5 & 2 \\ 4 & 1 & -8 & 6 \end{array}\right] \xrightarrow[R_3\to R_3-4R_1]{R_2\to R_2-3R_1} \left[\begin{array}{ccc|c} 1 & 2 & -3 & 4 \\ 0 & -7 & 4 & -10 \\ 0 & -7 & 4 & -10 \end{array}\right] \xrightarrow{R_3-R_2} \left[\begin{array}{ccc|c} 1 & 2 & -3 & 4 \\ 0 & -7 & 4 & -10 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

Now it is in echelon form

$\text{rank}(A_1)=2$, $\text{rank}(A_1|B_1)=2$, $n=3$

Because $\text{rank}(A_1)=\text{rank}(A_1|B_1)$ and $r<m$, the system is consistent and has $\boxed{\text{infinitely many solutions for } k=1}$

$k=-1$

$4x+y+(k^2-9)z=k+5 \Rightarrow 4x+y-8z=4$

$$\begin{cases} x+2y-3z=4 \\ 3x-y-5z=2 \\ 4x+y-8z=4 \end{cases} \qquad Ax=b \Rightarrow \begin{pmatrix} 1 & 2 & -3 \\ 3 & -1 & -5 \\ 4 & 1 & -8 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 4 \\ 2 \\ 4 \end{pmatrix}$$

$$[A_{-1}|B_{-1}] = \left[\begin{array}{ccc|c} 1 & 2 & -3 & 4 \\ 3 & -1 & -5 & 2 \\ 4 & 1 & -8 & 4 \end{array}\right] \xrightarrow[R_3 \to R_3 - 4R_1]{R_2 \to R_2 - 3R_1} \left[\begin{array}{ccc|c} 1 & 2 & -3 & 4 \\ 0 & -7 & 4 & -10 \\ 0 & -7 & 4 & -12 \end{array}\right] \xrightarrow[R_3 - R_2]{R_3 \to} \left[\begin{array}{ccc|c} 1 & 2 & -3 & 4 \\ 0 & -7 & 4 & -10 \\ 0 & 0 & 0 & -2 \end{array}\right]$$

$\text{rank}(A_{-1}) = 2$, $\text{rank}(A_{-1}|B_{-1}) = 3$, $n=3$

Because $\text{rank}(A_{-1}) < \text{rank}(A_{-1}|B_{-1})$, the system is not consistent and has $\boxed{\text{no solutions for } k=-1}$

$k=2$

$4x+y+(k^2-9)z=k+5 \Rightarrow 4x+y-5z=7$

$$\begin{cases} x+2y-3z=4 \\ 3x-y-5z=2 \\ 4x+y-5z=7 \end{cases} \qquad A_2 = \begin{vmatrix} 1 & 2 & -3 \\ 3 & -1 & -5 \\ 4 & 1 & -5 \end{vmatrix}, \quad B_2 = \begin{vmatrix} 4 \\ 2 \\ 7 \end{vmatrix}$$

$$\det(A_2) = 1(-1(-5)-1(-5)) - 2(3(-5)-4(-5)) + (-3)(3\cdot 1 - 4(-1)) = -21 \neq 0$$

$$\det(A_{2x}) = \begin{vmatrix} 4 & 2 & -3 \\ 2 & -1 & -5 \\ 7 & 1 & -5 \end{vmatrix} = 4(-1(-5)-1(-5)) - 2(2(-5)-7(-5)) + (-3(2\cdot 1 - 7(-1))) = -37 \neq 0$$

$$\det(A_{2y}) = \begin{vmatrix} 1 & 4 & -3 \\ 3 & 2 & -5 \\ 4 & 7 & -5 \end{vmatrix} = 1(2(-5)-7(-5)) - 4(3(-5)-4(-5)) + (-3)(3\cdot 7 - 4\cdot 2) = -34 \neq 0$$

$$\det(A_{2z}) = \begin{vmatrix} 1 & 2 & 4 \\ 3 & -1 & 2 \\ 4 & 1 & 7 \end{vmatrix} = 1(-1\cdot 7 - 1\cdot 2) - 2(3\cdot 7 - 4\cdot 2) + 4(3\cdot 1 - 4(-1)) = -7 \neq 0$$

$$x = \frac{\det(A_{2x})}{\det(A_2)} = \frac{-37}{-21} = \frac{37}{21} = 1.76 \qquad y = \frac{\det(A_{2y})}{\det(A_2)} = \frac{-34}{-21} = \frac{34}{21} \qquad z = \frac{\det(A_{2z})}{\det(A_2)} = \frac{-7}{-21} = \frac{1}{3}$$

Answers:

a) $k=-1$

b) $k=1$

c) $k \neq \pm 1$; for $k=2$: $x=\frac{37}{21}$, $y=\frac{34}{21}$, $z=\frac{1}{3}$

Problem 10

$$A=\begin{pmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & 3 & 4 \\ 2 & 3 & 1 & 0 \end{pmatrix} \qquad \Phi: \mathbb{R}_4 \to \mathbb{R}_3,\ x \to Ax$$

a) Let's find the echelon form for a see the rank of A.

$$\begin{vmatrix} 1^{2} & 0^{0} & 2^{4} & 1^{2} \\ 0 & 1 & 3 & 4 \\ 2 & 3 & 1 & 0 \end{vmatrix} \xrightarrow{R_3 \to R_3 - 2R_1} \begin{vmatrix} 1 & 0 & 2 & 1 \\ 0^{0} & 1^{3} & 3^{9} & 4^{12} \\ 0 & 3 & -3 & -2 \end{vmatrix} \xrightarrow{R_3 \to R_3 - 3R_2} \begin{vmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & 3 & 4 \\ 0 & 0 & -12 & -14 \end{vmatrix} \begin{matrix} R_1 \\ R_2 \\ R_3 \end{matrix}$$

$R_1 \neq 0$, $R_2 \neq 0$, $R_3 \neq 0$

$\boxed{\text{rank}(A)=3}$

b) Find RREF from REF

$$\begin{vmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & 3 & 4 \\ 0 & 0 & -12 & -14 \end{vmatrix} \xrightarrow{R_3 \to R_3 (-1/2)} \begin{vmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & 3 & 4 \\ 0^{0} & 0^{0} & 1^{2} & \frac{7}{6}^{3} \end{vmatrix} \xrightarrow{R_1 \to R_1 - 2R_3} \begin{vmatrix} 1 & 0 & 0 & -\frac{4}{3} \\ 0 & 1 & 3 & 4 \\ 0^{0} & 0^{0} & 1^{3} & \frac{7}{6}^{\frac{21}{6}} \end{vmatrix} \xrightarrow{R_2 \to R_2 - 3R_3}$$

$$\to \begin{vmatrix} 1 & 0 & 0 & -\frac{4}{3} \\ 0 & 1 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & \frac{7}{6} \end{vmatrix}$$

$$\text{rref}(A)=\begin{pmatrix} 1 & 0 & 0 & -\frac{4}{3} \\ 0 & 1 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & \frac{7}{6} \end{pmatrix}$$

c)

The image of mapping Φ is the span of pivot columns of A (c_1, c_2, c_3)

$$c_1 = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}, \quad c_2 = \begin{pmatrix} 0 \\ 1 \\ 3 \end{pmatrix}, \quad c_3 = \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix}$$

$$M = (c_1, c_2, c_3) = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 2 & 3 & 1 \end{pmatrix}$$

$$\det(M) = 1(1 \cdot 1 - 3 \cdot 3) - 0(0 \cdot 1 - 2 \cdot 3) + 2(0 \cdot 3 - 2 \cdot 1) = -12 \neq 0$$

$\operatorname{span}(M) = \operatorname{span}(c_1, c_2, c_3) = \mathbb{R}_3$

$\operatorname{Im}(\Phi) = \mathbb{R}_3$

Basis will be $\{c_1, c_2, c_3\}$ or $\{(1,0,2), (0,1,3), (2,3,1)\}$

d)

Let's solve $Ax = 0$ using $\operatorname{rref}(A)$

$$\begin{cases} x_1 - \frac{4}{3}x_4 = 0 \\ x_2 + \frac{1}{2}x_4 = 0 \\ x_3 + \frac{7}{6}x_4 = 0 \end{cases} \Rightarrow \begin{cases} x_1 = \frac{4}{3}x_4 \\ x_2 = -\frac{1}{2}x_4 \\ x_3 = -\frac{7}{6}x_4 \end{cases}$$

Let $x_4 = t \in \mathbb{R}$, then $x = t \begin{pmatrix} \frac{4}{3} \\ -\frac{1}{2} \\ -\frac{7}{6} \\ 1 \end{pmatrix}^{\times 6} = t \begin{pmatrix} 8 \\ -3 \\ -7 \\ 6 \end{pmatrix}$

$\ker(\Phi) = \operatorname{span}\{(8, -3, -7, 6)\}$